Fairness in Data Distribution: A Critical Lens on Disaster Response

D4F Assignment

Use Case Scenario: Fair Data Management in Disaster Response

Context: In disaster response scenarios, the integration of edge computing and cloud services plays a pivotal role in managing and processing the vast amounts of data generated during emergencies. Edge computing enables data processing closer to the source, facilitating real-time decision-making essential for effective disaster management (Xu et al. 2020). However, disparities in infrastructure and resource availability can lead to unequal access to these advanced technologies. Large organizations often possess the means to deploy sophisticated edge-computing devices and maintain robust cloud services, ensuring swift data processing and response times. In contrast, smaller agencies or those in resource-constrained regions may lack such capabilities, resulting in delays and inefficiencies in disaster response efforts (Musa et al. 2024).

Fairness Issue: The current technological landscape in disaster management tends to favor well-resourced organizations, leading to inequities in response capabilities. Smaller agencies or those operating in low-income communities often face challenges in adopting disruptive technologies like AI, machine learning, and robotics due to financial and infrastructural constraints (Musa et al. 2024). This disparity hampers their ability to respond effectively and promptly to disasters, exacerbating the vulnerabilities of the communities they serve. Moreover, the lack of standardized data management practices can lead to inconsistencies and hinder collaboration among different agencies during disaster response efforts.

Proposed Fairness-Oriented System Feature: To address these disparities, it is essential to develop equitable disaster response systems that leverage disruptive technologies while being accessible to all agencies regardless of their size or resources. Implementing standardized data management practices, such as the FAIR (Findable, Accessible, Interoperable, Reusable) principles can enhance data sharing and collaboration among agencies (Wilkinson et al. 2016). Additionally, investing in affordable and scalable edge computing solutions can empower smaller agencies to process data locally, reducing dependency on centralized cloud services and improving response times. Collaborative efforts and funding mechanisms should be established to support the adoption of these technologies in under-resourced communities, ensuring a more equitable and effective disaster response framework.

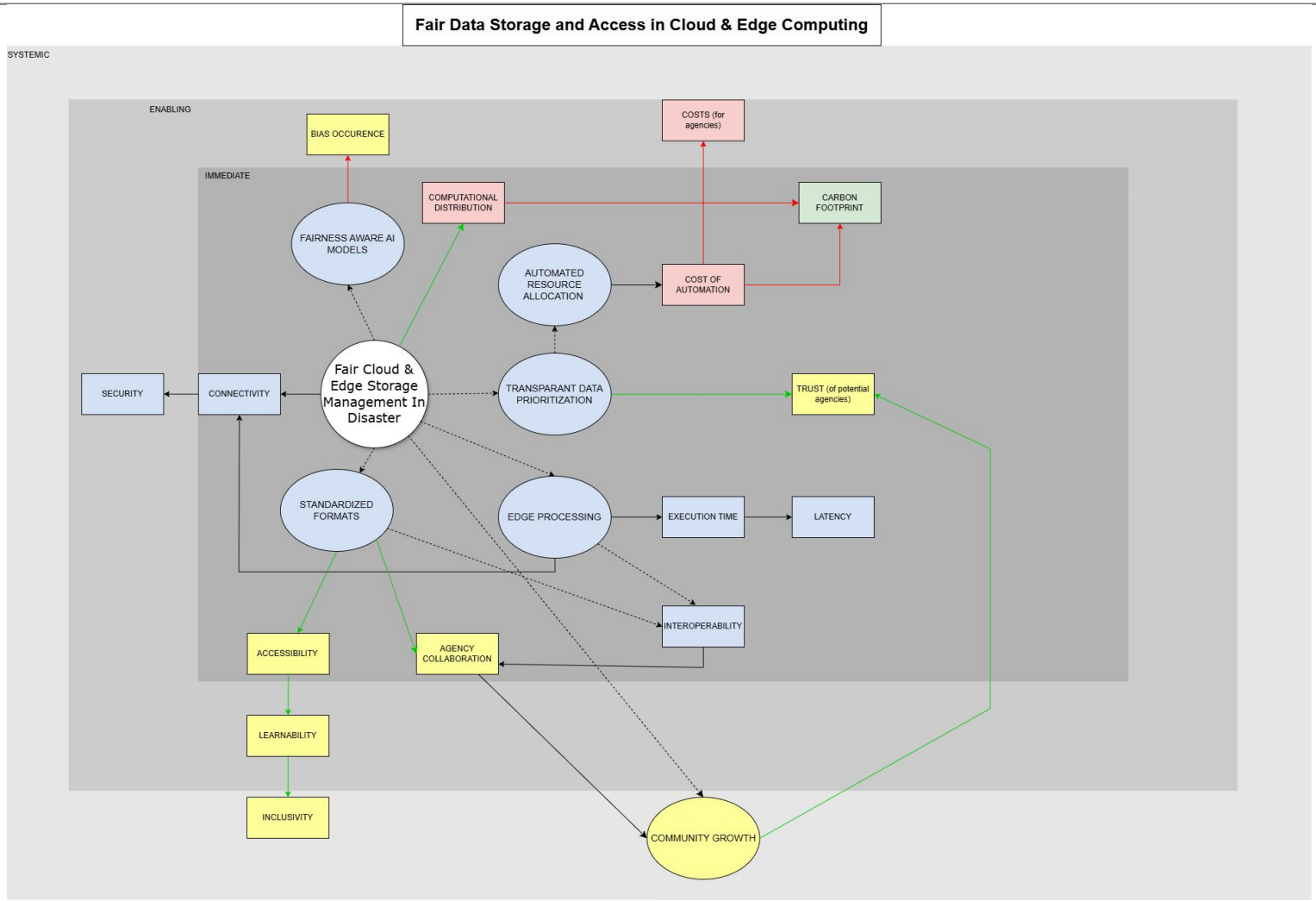
References

Xu, J., Ota, K., & Dong, M. (2020). Big data on the fly: UAV-mounted mobile edge computing for disaster management. *IEEE Transactions on Network Science and Engineering*, 7(4), 2620-2630. 10.1109/TNSE.2020.3016569

Musa, C. C., Liu, L., & Egiebor, M. (2024). Leveraging disruptive technologies for faster and more efficient disaster response: A focus on low-income communities. *Sustainability*, 16(23), 10730. **10.3390/su16231073**

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., et al. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3, 160018.1 https://doi.org/10.1038/sdata.2016.18

Definition of Fairness: Decision Map



Definition of Fairness: Decision Map (cont.)

Definition of Fairness: Fairness in cloud and edge storage means ensuring that all organizations regardless of size, financial power, or geographic location have equitable access to data processing, AI decision-making, and computing resources. It prevents systemic biases in disaster response by enabling transparent, explainable, and accountable data management while ensuring inclusivity and sustainable operations. A fair system must dynamically balance hot (real-time) and cold (historical) data, allocate computing power based on urgency rather than payment, and prevent monopolization by large entities.

System Features:

Edge Processing: Enables local computation to reduce latency and cloud dependency.
Automated Resource Allocation: Dynamically distributes compute power and bandwidth based on urgency.
Transparent Data Prioritization: Routes and processes data fairly, based on criticality rather than agency status.
Standardized Formats: Ensures interoperability between different systems and agencies.
Fairness-Aware AI Models: Prevents algorithmic bias and promotes explainability in AI decision-making.

Concerns: **Bias**
Occurrence: AI models may favor well-documented or high-resource regions.
Computational Distribution: Traffic towards cloud gets lessen as edge usage will take a part of that traffic to reduce dependency.
Costs: High financial barriers can prevent smaller agencies from adopting cloud and edge technologies.
Cost of Automation: Automation requires infrastructure, licenses or expertise in which may lack in some organizations.
Carbon Footprint: Increased computation that leads to higher energy consumption.
Execution Time: Slow data processing can result in critical delays during disaster, especially when relying on under-optimized systems.
Latency: Network lag between data exchanges may hinder real-time decision-making in time-sensitive scenarios
Interoperability (FAIR): Without standardized formats, systems from agencies may be unable to exchange data causing fragmentation.
Agency/Enterprise Collaboration: A lack of shared standards and best practices between enterprises/agencies may hinder effective cooperation during disasters.
Accessibility: Technological solutions may not be usable by all if they require high-end devices, stable internet, or prior digital literacy.
Learnability: Complex documented interfaces make systems difficult to adopt especially for first responders with limited time/knowledge.
Connectivity: Unstable or unavailable networks can render cloud-dependent services useless, especially in remote or disaster-struck areas.
Security: Inadequate protection of sensitive disaster data may lead to misuse, breaches, or non-compliance.

Definition of Fairness: SQ model

QUALITY ATTRIBUTE (QA)	DEFINITION	REFERENCE	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL	DEFINITION of METRIC/KPI
Execution Time (ET)	Time to process disaster data (in seconds).	ISO/IEC 25010:2011 (Performance Efficiency – Time behavior), "https://www.iso.org/obp/ui/#iso:std:iso-iec:2	X				ET = T(task) - T(baseline)
Latency (L)	Delay between user request and system response (in ms).	ISO/IEC TR 29181-3:2013 (Future Networks – Functional architecture), "https://www.iso.org/home.html"	X				L = Response_time - Request_time
Carbon Footprint (CF)	The total greenhouse gas emisssions resulting from computations and transmissions.	ISO 14001:2015, "https://www.iso.org/standard/60857.html"		X			CF = Energy_use × Emission_factor
Cost of Automation (CA)	Cost of implementing and maintaining automated processes (€ per year).	Cost Automation, https://www.sciencedirect.com/topics/engineering/cost-automation			X		CA = (Setup_cost + Maintenance_cost) / Year
Trust (T)	Confidence that the system behaves reliably, securely, and fairly.	ISO/IEC 25010:2011 (Quality in Use – Trust), "https://www.iso.org/obp/ui/#iso:std:iso-iec:2				X	TS (Trust Score) = (% of users expressing high trust)
Accessibility (A)	Extent to which the system is usable by people with different abilities.	Web Content Accessibility, "https://www.w3.org/TR/WCAG21/"	X				A = (# accessible elements / # total elements) × 100
Interoperability (IO)	System's ability to use and exchange data with other systems.	ICT standards for Procurement, "https://interoperable-europe.ec.europa.eu/collection/ict-standards-procurement/interop	X				IO = (# compatible interfaces / # total interfaces) × 100
Security (S)	Degree to which system protects data from unauthorized access or modification.	ISO/IEC 27010:2015 (Information security for inter-agency communication), "https://www.iso.org/standard/68427.html"	X				S = (# incidents_prevented / # attempts) × 100
Connectivity (C)	Availability of reliable communication between system components or to the cloud.	ISO/IEC 23005-1:2016 (Network Interconnectivity), "https://www.iso.org/obp/ui/#iso:std:iso-iec:2	X				Uptime = (Available_time / Total_time) × 100
Agency Collaboration (AC)	Effectiveness of cooperation and coordination across agencies.	ACM Digital Government & Crisis Informatics, "https://dl.acm.org/doi/10.1007/s10606-014				X	Collab_score = (# shared_actions / # total_actions) × 100
Learnability (L)	Ease with which users can learn how to use the system.	ISO/IEC 25010:2011 (Usability – Learnability), "https://www.iso.org/obp/ui/#iso:std:iso-iec:2				X	Learn_time = Avg. time to complete key tasks
Inclusivity (I)	Extent to which diverse user groups can effectively use the system.	ISO/IEC 25010:2011 (Usability – Accessibility/Universality), "https://www.iso.org/obp/ui/#iso:std:iso-iec:2				X	Inclusivity = (# supported profiles / # total profiles) × 100
Costs (CO)	Total cost of ownership, including system deployment, use, maintenance, and support.	ISO/IEC 19770-1:2017, "https://www.iso.org/obp/ui/en/#iso:std:iso-iec:19770:-1:ed-3:v1:en"			X		TCO = Capex + Opex
Bias Occurrence (BS)	The extent to which system outputs systematically favor or disadvantage particular groups or contexts.	ISO/IEC TR 24027:2021, "https://www.iso.org/standard/77607.html"				X	Fairness Score = 1 - (Accuracy_Agency_A - Accuracy_Agenc
Computation Distribution (CD)	The degree to which computational load is balanced between edge and cloud resources to ensure scalability, resilience, and fairness	ISO/IEC 30141:2018 (Reference architecture for IoT) + IEEE Edge Computing Taxonomy			X		Edge_ratio = (Edge_compute / Total_compute) × 100 Cloud_ratio = (Cloud_compute / Total_compute) × 100 Balance_index = (Edge_compute - Cloud_compute) / Total_compute

Definition of Fairness: SQ model (cont.)

The SQ model identifies and defines key sustainability-related quality attributes, each measurable through concrete metrics and KPIs. While these attributes span technical, economic, environmental, and social dimensions, their shared foundation is the principle of fairness. In this context, fairness refers to the equitable access to system resources and benefits, regardless of the stakeholder's institutional size, financial capacity, or geographic location. This interpretation aligns with the fairness principles outlined in ISO/IEC TR 24028:2020.

A fair system ensures that critical computational power is not monopolized by well-funded actors but is distributed based on need and urgency (Singh & Chana, 2016). It also ensures equal access to data dependable on the severity of the disaster and system functionalities through interoperable, accessible, and inclusive design, as recommended in ISO/IEC 25010:2011 and ISO/IEC 40500:2012. The SQ model operationalizes this by including both **direct fairness attributes** such as *Accessibility*, *Bias Occurrence*, *Inclusivity*, and *Trust* and **indirect ones**, such as *Execution Time*, *Costs*, and *Interoperability*, which influence fairness through systemic efficiency and openness.

By measuring these attributes using standardized metrics and ensuring their balance across sustainability dimensions, the model enables architects to evaluate whether the system supports equity, transparency, and accountable performance. This is particularly relevant in socio-technical systems like disaster response, where fairness is not just ethical but operationally critical.

References:

ISO/IEC 25010:2011. *Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE)*
ISO/IEC 40500:2012. *Information technology – W3C Web Content Accessibility Guidelines (WCAG) 2.0*
ISO/IEC TR 24028:2020. *Overview of trustworthiness in AI systems*
Singh, S., & Chana, I. (2016). *A systematic review of cloud cost estimation techniques*. ACM Computing Surveys (CSUR), 49(1), 1–38. https://doi.org/10.1145/2851196

Fairness perspective

	DESCRIPTION
DESIRED QUALITY	*Ensuring fair and sustainable data management where data accessibility, processing, and storage do not disproportionately benefit large enterprises at the expense of smaller businesses or individual users. The system must balance energy efficiency and equitable resource distribution.*
CONCERNS	***Data Access Equity:*** *Ensuring fair access to both hot and cold data for all businesses, regardless of size.* ***Bias-Free Processing****: Preventing algorithmic bias in data retrieval and prioritization.* ***Energy-Efficient Fairness****: Maintaining low energy consumption while ensuring equitable access to data.* ***Regulatory Compliance****: Aligning with GDPR, SOX, and ISO standards to ensure transparent and ethical data handling.* ***Transparent Cost Models****: Avoiding pricing models that favor large enterprises over SMEs.*
ARCHITECTURAL TACTIC(S)	***Smart Data Tiering****: Implement intelligent caching and prefetching mechanisms to balance latency and storage costs while maintaining fairness.* ***Adaptive Resource Allocation****: Use dynamic workload distribution algorithms to optimize computing resource allocation based on real-time demand.* ***Green Computing Strategies:*** *Implement low-energy-consumption architectures in cloud data centers and edge nodes to reduce carbon footprint.* ***Explainable AI for Decision-Making****: Ensure AI-driven data access and processing decisions are interpretable and justifiable to avoid bias.* ***Regulated Storage Access:*** *Implement compliance driven retention policies that guarantee audit-ready data access across all user groups.*
POSSIBLE KPI(S)	*Fairness Index (FI): FI = (Average Access Time (SME)/Average Access Time (Enterprise)) * 100* *Bias Detection Rate (BDR): (Detected Biases in AI Decisions/Total AI Decisions) * 100.* *Energy Efficiency Improvement (EEI): EEI = (Baseline Energy Usage - Current Energy Usage)/Baseline Energy Usage) * 100* *Regulatory Compliance Score (RCS): RCS = (Compliant Data Requests/Total Data Requests) * 100*
PROBLEMS AND PITFALLS	***Unintended Bias in AI Processing****: Even with fairness mechanisms, unbalanced training datasets may introduce unintended bias.* ***High Energy Consumption for Real-Time Access****: Increasing data accessibility in a fair manner may increase computational load, conflicting with sustainability goals.* ***Regulatory Complexity****: Meeting multiple jurisdictional regulations (e.g., GDPR, SOX, ISO 27001) while maintaining fairness may require frequent policy adjustments.* ***Scalability Concerns:*** *Implementing equitable resource distribution at scale may introduce bottlenecks in large-scale cloud-edge environments.* ***Cost vs. Fairness Trade-Off:*** *Some pricing models favor higher-paying customers, making it difficult to implement a fully equitable pricing structure without economic drawbacks.*

Definition of KPI*

KPI Name					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Ensure **equal data access** between small businesses and large enterprises in a cloud-edge hybrid system.	Minimizing latency disparity between SMEs and large enterprises in accessing both **hot and cold data**.	The **Fairness Index (FI)** measures the relative access time of small businesses compared to large enterprises.		**Formula:** Fairness Index (FI): FI = (Average Access Time of Small Business Enterprise/Average Access Time of Large Business Enterprise) * 100 If **FI = 100%**, access is perfectly fair. If **FI < 100%**, small businesses experience slower access times, indicating unfairness.	**Data Collection Frequency:** Monthly **Unit of Measure:** Percentage of Access Time from both Small- & Large Enterprises (%) **Threshold:** FI should not drop below **90%**
		Target	**Action(s)**		
		Achieve a Fairness Index of **≥ 95%**.	Implement smart caching mechanisms to optimize data retrieval speed. Deploy resource allocation adjustments to balance workloads dynamically. Conduct bias detection in AI-based access		

Architectural Tactic(s) for Fairness* (½)

Architectural Tactic: Reducing Latency in Flood Disaster Response Using Edge Computing

TARGET QA: Latency in Flood Disaster Response Systems
TACTIC INTENT: Reduce latency in flood monitoring and response coordination by enabling real-time data processing at the edge, minimizing delays in flood prediction, rescue operations, and resource deployment.
CONTEXT: Drones and satellite imagery provide supplementary flood progression tracking and damage assessments. Large disaster agencies with advanced cloud computing can process this data quickly, while smaller, under-resourced local emergency units often face latency delays due to poor connectivity and centralized data processing constraints.
PARTICIPANT: Local Emergency Response Teams (e.g., Firefighters, Civil Defense, Municipal Authorities) rely on low-latency flood data to coordinate evacuations and deploy rescue efforts.
ARTIFACT: Edge-Based Flood Monitoring System: Deploys edge nodes at critical flood-prone locations to process water level sensor data locally & Uses AI-driven predictive analytics at the edge to detect flood risks before sending data to the cloud.
SOFTWARE FEATURE: Localized Flood Risk Prediction Model: runs on edge devices enabling immediate anomaly detection in water level fluctuations & FAIR-Compliant Data Sharing Framework; Standardizes sensor data formats across different agencies to ensure seamless information flow between local and national response teams.
OTHER RELATED QAs:

- **Interoperability:** Ensures cross-agency collaboration by enabling standardized data exchange between meteorological institutes, emergency responders, and local governments.
- **Security**: Protects real-time flood sensor data from cyber threats, ensuring reliability in emergency response systems.
- **Connectivity**: Ensures uninterrupted communication between edge nodes, mobile response teams, and centralized cloud servers.

References:
VU Fair Digital Design Reference Material, 2025
Karaman, B., Basturk, I., Taskin, S., Zeydan, E., Kara, F., et al. (2024). *Solutions for sustainable and resilient communication infrastructure in disaster relief and management scenarios*. **ArXiv preprint arXiv:2410.13977**.

Architectural Tactic(s) for Fairness* (2/2)

Measured Impact

Flood Response Latency Index (FRLI)

The FRLI quantifies the effectiveness of edge computing in reducing response time during flood emergencies. It is calculated as:

FRLI = (Average Data Processing Time (Cloud-Based) / Average Data Processing Time (Edge-Based)) × 100

- FRLI > 100% indicates that edge computing significantly reduces latency, enabling faster detection and response.
- FRLI ≈ 100% suggests negligible improvement, potentially pointing to under-optimized edge configurations.

Evacuation Alert Efficiency (EAE)

EAE measures how effectively the system alerts and evacuates people before the flood peak is reached. It is defined as:

EAE = (People Evacuated Before Flood Peak / Total Affected Population) × 100

A higher EAE indicates that a larger proportion of the at-risk population received and responded to alerts in time, reflecting both communication efficiency and user trust in the system.

Network Uptime in Disaster Zones (NUDZ)

NUDZ captures the system's ability to maintain connectivity for real-time monitoring during disaster conditions. It is expressed as:

NUDZ = (Time Edge Devices Maintain Connectivity / Total Disaster Duration) × 100

A high NUDZ value demonstrates the resilience of edge infrastructure and its critical role in continuous monitoring, especially when traditional networks are disrupted.

References:

PreventionWeb/UNDRR. (2022). *Saving lives with FAIR data: Evidence-based public policies and the impact of data in disaster and climate change risk management*. United Nations Office for Disaster Risk Reduction. (Retrieved from Climate Adaptation Platform summary, Nov 2022)

Fairness view: Fair & Efficient Flood Disaster Response Using Edge Computing

Fairness-Oriented Flood Disaster Response: Reducing Latency with Edge Computing

Problem: Unequal Access to Real-Time Flood Data

- Large agencies process data instantly via **advanced cloud computing**.
- Smaller, under-resourced teams **face latency delays**, slowing evacuation and response.

Solution: Edge-Based Flood Monitoring System

- **Deploys edge nodes** at flood-prone locations for **local data processing**.
- **AI-driven flood risk prediction** ensures **immediate alerts**.
- **FAIR-compliant data sharing** enables **seamless collaboration across agencies**.

Key Architectural Viewpoints Applied

- **Context Viewpoint** – Ensures **smaller teams receive real-time flood alerts**.
- **Information Viewpoint** – **Standardizes data formats** for better coordination.
- **Deployment Viewpoint** – Expands **edge infrastructure** for **equal computing access**.
- **Availability & Resilience Perspective** – Introduces **failover systems** to **prevent downtime**.

Key Metrics for Fairness & Sustainability

- **FRLI > 100%** → Confirms **faster flood response via edge computing**.
- **High EAE** → Ensures **equal access to evacuation alerts**.
- **NUDZ > 95%** → Guarantees **continuous monitoring during disasters**.

References:

Silva, A. F. R., & Eleutério, J. C. (2023). *Analysis of flood warning and evacuation efficiency by comparing damage and life-loss estimates with real consequences related to the São Francisco tailings dam failure in Brazil*. **Natural Hazards and Earth System Sciences, 23**, 3095–3110. https://doi.org/10.5194/nhess-23-3095-2023

Fairness view (cont.)

Ensuring Fair & Equitable Access to Flood Data

How This Solution Promotes Fairness:

- Eliminates dependency on cloud computing, ensuring equal response times.
- FAIR-compliant data sharing enhances collaboration across agencies.
- Edge computing minimizes latency, enabling faster decision-making for all teams.

Performance Indicators:

1. **FRLI** – Measures latency reduction with edge computing.
2. **EAE** – Tracks how many people evacuate before peak flooding.
3. **NUDZ** – Ensures real-time monitoring stays operational.

Challenges & Solutions:

- **Limited infrastructure → Deploy affordable edge devices & mobile units (UAVs)**.
- **Data security concerns → Implement encryption & access controls**.
- **Training needs → Provide automated updates & user-friendly tools**.

Conclusion: By reducing latency and standardizing data sharing, edge computing ensures fair, efficient, and sustainable flood disaster response for all communities, regardless of resources.

References:

Yang, T.-H., Wang, C.-W., & Lin, S.-J. (2020). *ECOMSNet – An edge computing-based sensory network for real-time water level prediction and correction*. **Environmental Modelling & Software, 131**, 104771. https://doi.org/10.1016/j.envsoft.2020.104771